

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **RTL modules in SystemVerilog** for high-speed digital signal processing pipelines on HCL's Ethernet infrastructure projects, achieving **timing closure** at multi-gigahertz clock rates and supporting **10Gb through 400Gb** throughput architectures.
- Developed **state machine logic** and **clocking architectures** for production Ethernet MAC and PCS controllers, collaborating with verification engineers to validate **digital design best practices** across multi-rate, multi-port configurations using **Xcelium and VCS simulation tools**.
- Performed **synthesis and timing analysis** on configurable IP cores using industry-standard EDA tools, closing timing violations and optimizing area-delay tradeoffs; applied **FPGA implementation** techniques to prototype and validate RTL behavior pre-silicon.
- Executed production support and reliability engineering for mission-critical Ethernet and networking systems, resolving **P1/P2 incidents** with **sub-minute SLA response times** and implementing automated monitoring dashboards using **CloudWatch and Dynatrace** to detect signal integrity and FEC anomalies.
- Collaborated cross-functionally with design, verification, and back-end engineering teams in co-located environments; participated in daily standups, design reviews, and integration testing to ensure seamless handoff from RTL through silicon delivery.
- Applied **digital design fundamentals** including **metastability analysis, clock domain crossing (CDC), and FEC error-correction concepts** to solve real-world timing and signal integrity challenges in high-speed networking IP; documented design decisions and test results in technical specifications.

SKILLS

RTL & Digital Design – SystemVerilog, Verilog, state machines, clocking architecture, timing closure, synthesis, FPGA implementation.

EDA & Simulation Tools – Xcelium, VCS, ModelSim, Vivado, timing analysis, constraint definition, place-and-route.

High-Speed Networking & Standards – Ethernet MAC, PCS, FEC, IEEE 802.3, multi-rate multi-port architectures, signal integrity, 10Gb–1.6Tb IP cores.

System Design & Collaboration – cross-functional team engineering, design reviews, verification coordination, silicon delivery lifecycle, co-located agile workflows.

Problem-Solving & Analysis – analytical debugging, timing violation root-cause analysis, clock domain crossing, metastability mitigation, performance optimization.

Production Support & Reliability – incident response, SLA management, operational monitoring, CloudWatch, Dynatrace, system observability.

Software & Data Engineering – Python, Java, SQL, AWS, Docker, CI/CD, Git, Jenkins, data pipeline design.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

High-Speed Ethernet MAC & PCS RTL Design (HCL Technologies)

- Authored **SystemVerilog RTL** for **Ethernet MAC and PCS controller cores** supporting **10Gb, 25Gb, 100Gb, and 400Gb** multi-rate operation, implementing **state machines, clock domain crossing, and configurable port architectures** to handle diverse network topologies.
- Performed **synthesis and timing closure** using industry EDA tools, resolving critical-path violations and meeting sub-nanosecond timing budgets; coordinated with back-end engineers on place-and-route to achieve **1.6Tb aggregate throughput** specifications.
- Integrated **FEC (Forward Error Correction) logic** and **signal integrity constraints** into RTL design; validated behavior across process, voltage, and temperature (PVT) corners using **Xcelium and VCS simulation** to ensure production-grade reliability.

Ethernet Standards & IEEE 802.3 Compliance Validation (HCL Technologies)

- Analyzed **IEEE 802.3 and Ultra Ethernet Consortium specifications** for Ethernet MAC, PCS, and FEC standards; translated protocol requirements into **RTL design specifications** and test vectors to ensure standards-compliant behavior across **10Gb through 1.6Tb** speed grades.
- Designed comprehensive **digital design verification testbenches** in SystemVerilog, validating **clocking, timing, state machine transitions, and error-correction functionality** against protocol specifications; achieved **95%+ code coverage** and zero protocol-compliance failures in pre-silicon validation.
- Collaborated with verification and standards-committee engineers to debug edge cases in high-speed packet processing; documented design trade-offs and signal integrity trade-offs in technical reviews, enabling downstream silicon team to meet **timing closure and yield targets**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)